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Purpose/Policy:

This standard operating procedure will help guide and instruct operation, changeover, troubleshooting and, cleaning of the Switchback Tray Packer. This SOP will include multiple changeover formats into the Loose tray format.

Scope:

The Switchback Tray Packer's purpose is to put loose cans into a 24 pack tray.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- E-Stop – Emergency Stop button
- Cartoner – Machine that packages cans into wraps (cartons)
- Switchback Tray Packer – Machine that puts wraps, Paktech units or loose cans into trays, or passes wrapped cases through

Authorities/Responsibilities:

This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to assist with "Loose Tray Changeover". A trainer will be responsible for showing how to make the changeover adjustments in correlation with this SOP document.

Resources and Materials:

Operations:

- Air.
- Trays.
- Loose cans.

Changeovers:

- One Tray (for setting flights).
- Spacers for compression pads.
- Pusher pad for Transfer area.

Tools:

- 1/2 “Wrench (or 13mm).
- 7/16” Wrench.
- 9/16” Socket.
- 3/16” Hex Wrench.

Basic Cleaning:

- Alpha 5.
- Warm Water.
- Dry Rags.
- Shop Towels.
- Glue Scraper.

Deep Clean:

- Refer to Zone Cleaning SOP.
    - N:\Operations\Production\Boston\Packaging\Canning Line 2024\Zone Cleaning\Can Line Lower Level\Zone 7 - Lower Level - Cartoner, Tray Packer

Troubleshooting:

- No Equipment required.

PPE Required:

- No formal PPE required, suggested nitrile gloves for cleaning.

Procedures:

Changeover Guide:

Recipe Data:

1. From the HMI select “Recipe Data.”
2. Then select “Setup Data.”
3. Using the arrows select the corresponding Loose recipe (Recipe #04 24LSX12TRY).

Note: Do not load the recipe yet.

Adjusting the Magazine Deck:

1. Starting with #2 adjust the “Mag Back Guide.”
2. Reference #3 to set up the “Mag Front Guide.”
3. Reference #4 to set up the ”Mag Deck.”

4. Reference #5 to set up the “Mag Top Guide.
5. Reference #8 to set up the "Mag Right.”
6. Reference #9 to set up the "Mag Center Clip.”
7. Reference #10 to set up the "Mag Left Clip.”

Note: “Mag Long Right” is set out of the way unless double picking occurs.

Strongback Adjustment:

1. Set the Strongback to what #1 says on the HMI.
2. Once you have set the Strongback place the clear rubber cover on the outfeed axle.

Non-Load Side Changeover:

1. If changing over from Passthrough attach prebreak pad underneath the mag deck.
2. Adjust the guiderail for the major flaps so that they tuck underneath the rail. This should be at a shallow angle.
3. Adjust the guide plate so the minor flap can clear it when tucked but will also keep folded during transfer.
4. Lower the “Ski Jump” so that it is out of the way.
5. Remove all the spacers for the compression pads.
6. If coming from passthrough add in wooden space at the outfeed.
7. If coming from passthrough add in outfeed catch/plate.

Glue Height Adjustment:

1. Loosen the orange handle for the non-load side glue nozzle and adjust the height according to #12.
2. After setting the height tighten the orange handle back down.
3. Go over to the other side of the tray packer and loosen the load side orange handle.
4. Set the glue nozzle height according to #11 and then tighten down the orange handle.

Load Side Changeover:

1. Loosen the two ½” bolts on the guide rail next to glue assembly. and slide the rail all the way up towards the magazine.
2. Remove all the spacers for the compression pads.
3. Adjust the major flap guide towards the non-load side at a shallow angle so that it guides the flap underneath.
4. Remove the plate that blocks the loose infeed lanes.
5. Make sure to put the bolts that held the plate up back in to keep holding the can sensors.

6. If coming from passthrough loosen both orange handles on the transfer area back stop and move it out of the way. Tighten down the orange handles. Make sure it is in the extended position.
7. Using 3/16” hex key, remove the pusher pad and put in the pusher pad for loose.
8. Put the 24pk loose lane assembly into line by loosening the orange handles and sliding it towards the rack. Once in place tighten down the orange handles.

Loading the Program:

1. Make sure the Tray Packer is in test mode.
2. Then from the “Setup Data” screen with the corresponding packaging size. Then hit the “USE” button.
3. This will change to “CAUTN ACTV” and turn orange.
4. Next press the E-Stop button and then pull it back out.
5. Press the black reset button to send everything to their home positions.

Setting up and Operating the Tray Packer:

1. From the home screen, press the “Run Mode” button to engage packaging operations.
2. Turn on the Glue by pressing the power button twice on the Glue Unit HMI if the screen is not lit. If the screen is lit, press the power button once. Flashing green indicates that the unit is warming up and orange is in standby mode. Make sure the light on the glue machine HMI is solid green.
3. Load the correct size tray for the current packaging run.
4. From the HMI home screen, turn on the following:
    4.1 “Glue.”
    4.2 “Vacuum Pick.”
    4.3 “Magazine Conv. On.”
5. Inspect trays for uniformity, glue shot placement and structure. Adjust machine as needed.

Note: For basic adjustments and troubleshooting, please refer to the “Troubleshooting Guide” section of this SOP document.

Shutting Down the Tray Packer:

1. After the “push” has been completed by the Can Line Filler Operator, the Tray Packer will need to be “Packed out.”

Note: Please see below for instructions and refer to operator training on how to complete.

2. Try to shut off the “Magazine Vacuum” so that the appropriate number of trays are loaded into the machine for the remaining cases.

3. Rearrange cans evenly in lanes.
4. If too difficult or can't flag sensors but you have a full case, load the cans by hand, go into system status screen and press "transfer case load."
5. If an extra tray is picked stop the machine and open the door and remove it.
6. Once the extra tray is removed reset the tray packer and press the green start button.
7. Stop the machine and remove any partial cases that are in the sweep area and pre-transfer area as needed.

Basic Cleaning of the Tray Packer:

Note: A hose should never be used inside the Tray Packer and please note the water should never be sprayed near the control panels or HMI screens.

1. Using a spray bottle of "Alpha 5", spray the Tray Packer stainless steel areas and any transfer areas.
    1.1 Wipe areas with a dry rag until clear of beer build-up and dust accumulation.
2. Use a rag and "Alpha 5" to clean the plexiglass windows. Never use a scrub pad on plexiglass.
3. Use the air gun to blow out any cardboard dust present in or on the machine.

Deep Cleaning:

1. Refer to Zone Cleaning SOP.
    1.1 N:\Operations\Production\Boston\Packaging\Canning Line 2024\Zone Cleaning\Can Line Lower Level\Zone 7 - Lower Level - Cartoner, Tray Packer

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

1. Trays not going into pockets correctly.
    1.1 Start by observing where the pick arm is placing it.
    1.2 You are able to E-Stop the pick arm right as it's putting the tray in the pocket but before the tray vacuum come up to help see where it is going.
    1.3 If it's too far downstream you will want to adjust the magazine up to help with this issue.

1.4 If it's too far upstream you will want to adjust the magazine lower to help with this issue.

Note: Make small adjustments to prevent the trays from getting destroyed or falling into the machine.

2. Trays not getting picked by the pick arm.
   - 2.1 Start by checking the top guide and making small adjustments by either raising it or lowering it. You will need to check to see which direction is helping the situation.
   - 2.2 Check to make sure the trays themselves are not bowed.
   - 2.3 Check the pick arm filters to make sure they are not clogged with dust.
3. Major flap tearing.
   - 3.1 On the load side sometimes the major flap lifter/tucker doesn't bring the major flap up causing it to tear.
     - 3.1.1 Use a damp rag with some Alpha 5 to clean the plate and piston.
   - 3.2 Make sure the correct spaces are in for the compression pads.
   - 3.3 Make sure the major flaps are clearing the glue units.
     - 3.3.1 Please get maintenance assistance if major flaps are hitting the glue units.
4. Trays not gluing.
   - 4.1 Height adjustments to the glue nozzles might be necessary.
   - 4.2 Double check to make sure the glue units are firing.

Note: Check with your trainer or senior can line operator on how to test fire glue nozzles.

   - 4.3 Make sure the shots are actually hitting the tray and not hitting a rail.
   - 4.4 Double check to make sure a crash has not moved the sensor out of the way.
   - 4.5 Check to make sure the flights are square.
5. If gates aren't letting enough can through to the transfer area or are sending incomplete loads, double check and make sure all the sensors are clean
   - 5.1 It is also possible that the infeed lanes are sticky and will need to be cleaned.
6. Can conveyor faulting out.
   - 6.1 Run the product out and then hose off conveyor with hot water. Make sure to do this when it begins to fault as if left for too long, it will begin to fault out very frequently.
7. Trailing minor flap not folding or popping back out as it transfers.
   - 7.1 Make sure the guide plate is close to the minor flap tucker and the flap tucker clears with a small amount of room.

Note: This adjustment can be made easier by having a tray in the load position and manually tucking the flap in and setting the plate according to that.

Note: Make sure this adjustment doesn't block the sensor that detects the tray in the load position.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
001	2024/06/12		Kyle Szymanski
002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

Version Approval Number: 002

Approved by: Kyle Szymanski

Position Title: OSCI Manager

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